

ATIKOKAN, ON, Canada

WMO: 717470

Lat: 48.7610N

Lon: 91.6283W

Elev: 389

StdP: 96.73

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.5	-37.6	0.1	-33.6	-34.3	0.2	-30.4	7.2	-16.1	6.3	-14.4	0.6	250	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	29.0	20.3	27.4	19.1	25.8	18.1	21.9	26.9	20.8	25.4	19.7	23.7	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.7	24.6	19.1	14.6	23.2	18.1	13.7	22.1	66.2	27.0	62.0	25.6	58.1	23.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	5.0	DB	-37.1	32.0	4.0	1.6	-39.9	33.1	-42.3	34.1	-44.5	35.0	-47.4	36.2	(4)
(5)			WB	-37.0	23.8	3.9	1.5	-39.8	24.8	-42.1	25.7	-44.4	26.6	-47.2	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-15.4	-12.9	-5.2	2.8	10.2	16.0	18.5	17.3	12.8	5.2	-3.2	-11.3	(6)
(7)		DBStd	13.17	8.23	7.86	7.64	5.50	4.79	3.90	3.13	3.34	4.56	4.76	6.13	7.71	(7)
(8)		HDD10.0	3429	788	641	473	223	56	3	0	1	22	166	395	661	(8)
(9)		HDD18.3	5707	1047	874	731	467	255	90	37	61	173	408	645	919	(9)
(10)		CDD10.0	864	0	0	1	6	63	182	263	226	107	16	1	0	(10)
(11)		CDD18.3	101	0	0	0	0	4	19	42	28	8	0	0	0	(11)
(12)		CDH23.3	1054	0	0	0	2	64	224	409	274	78	3	0	0	(12)
(13)		CDH26.7	219	0	0	0	0	13	48	91	57	11	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.0	2.1	2.2	2.3	2.3	2.0	1.9	1.8	1.9	2.2	2.2	2.0	(14)
(15)	Precipitation	PrecAvg	797	36	26	40	51	76	108	106	102	91	80	48	37	(15)
(16)		PrecMax	1009	131	47	95	118	153	170	171	269	248	189	102	80	(16)
(17)		PrecMin	610	11	7	4	9	19	54	28	25	27	24	14	17	(17)
(18)		PrecStd	120	26	11	24	31	36	34	37	54	53	44	22	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	7.2	16.2	21.7	28.5	31.0	31.3	30.5	28.1	22.6	14.0	5.0	(19)
(20)			MCWB	0.6	3.2	10.4	11.6	17.9	22.2	21.5	21.1	19.6	15.3	9.7	2.3	(20)
(21)		2%	DB	0.7	3.5	11.0	18.1	25.1	27.7	29.0	28.2	25.6	19.0	10.6	2.2	(21)
(22)			MCWB	-0.2	1.1	6.3	9.2	15.2	18.7	20.4	20.2	18.6	13.4	7.8	0.7	(22)
(23)		5%	DB	-1.1	1.3	7.8	15.4	22.3	25.8	27.3	26.3	23.3	15.9	7.4	0.6	(23)
(24)			MCWB	-2.0	-0.2	4.0	7.8	13.5	17.5	19.5	19.1	17.4	11.7	5.0	-0.3	(24)
(25)		10%	DB	-3.4	-0.9	5.0	12.7	19.7	24.1	25.6	24.5	21.1	13.0	5.0	-0.7	(25)
(26)			MCWB	-4.1	-2.2	1.8	6.3	12.4	16.6	18.4	18.1	16.2	9.6	3.3	-1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	4.2	11.0	13.5	19.6	22.6	23.3	23.1	21.5	17.0	11.0	2.7	(27)
(28)			MCDWB	2.6	6.7	14.1	19.1	25.2	29.0	28.6	27.8	25.9	20.0	12.8	4.1	(28)
(29)		2%	WB	0.0	1.6	7.0	10.4	17.2	20.6	22.1	21.5	19.8	14.2	7.7	0.9	(29)
(30)			MCDWB	1.0	3.1	10.5	16.7	22.0	25.8	26.8	26.4	23.8	18.4	10.5	1.8	(30)
(31)		5%	WB	-1.9	-0.1	4.1	8.5	15.1	19.1	20.9	20.3	18.3	12.1	5.2	-0.1	(31)
(32)			MCDWB	-1.0	1.5	7.2	14.4	19.8	23.2	25.4	24.5	22.1	15.5	7.1	0.6	(32)
(33)		10%	WB	-4.0	-2.3	2.1	6.7	13.3	17.8	19.7	19.1	16.8	10.0	3.2	-1.5	(33)
(34)			MCDWB	-3.3	-1.0	4.8	12.4	18.6	22.0	23.7	23.1	20.4	12.4	5.0	-0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.0	12.7	12.8	12.4	12.5	11.8	11.7	11.8	11.0	8.8	7.4	9.1	(35)
(36)			MCDBR	9.9	12.6	14.0	17.4	17.0	14.6	14.0	14.0	14.0	13.9	11.4	9.0	(36)
(37)			MCWBR	8.7	10.0	9.3	10.1	8.6	7.0	6.7	7.0	7.9	8.8	8.2	7.6	(37)
(38)		5% WB	MCDBR	9.2	10.4	13.1	15.4	13.3	11.5	11.8	11.9	11.9	12.7	10.2	8.4	(38)
(39)			MCWBR	8.2	8.4	9.2	9.6	7.6	6.6	6.5	6.7	7.3	8.6	7.7	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.262	0.295	0.329	0.349	0.370	0.383	0.364	0.354	0.306	0.285	0.254	(40)	
(41)		taud	2.371	2.351	2.404	2.447	2.446	2.406	2.395	2.478	2.479	2.591	2.525	2.431	(41)	
(42)		Ebn at Noon	828	905	926	923	912	890	872	876	850	851	791	784	(42)	
(43)		Edh at Noon	65	85	96	103	108	113	113	99	90	68	58	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.22	2.41	3.70	4.80	5.27	5.60	5.78	4.93	3.57	2.09	1.12	0.88	(44)	
(45)		RadStd	0.18	0.27	0.32	0.47	0.31	0.45	0.34	0.39	0.29	0.28	0.12	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (21 neighbors)		(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A	(r) N/A	(s) N/A	(t) N/A	(u) N/A

Nomenclature: See separate page